

REPORT

ASSIGNMENT 7

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LOGIC BEHIND IMPLEMENTATION

1. **State Initialization:** The system starts in state 0 with dataValid set to 0, indicating the absence of valid input data.

2. **Input Data Handling:** Transitioning to state 2 upon receiving a dataValid signal of 1, the system begins processing input data. A counter is maintained to track input count, incrementing with each data input until dataValid returns to 0.

3. **Sorting Algorithm Implementation:** Implemented within the FSM, the Selection Sort algorithm operates at the **gate level**. Each FSM state corresponds to a specific step in the sorting process, manipulating counters for the outer and inner loops of the algorithm (i and j).

4. **Sorting Loop Termination:** The sorting process concludes when the outer loop counter (i) matches the total input count, indicating completion of sorting.

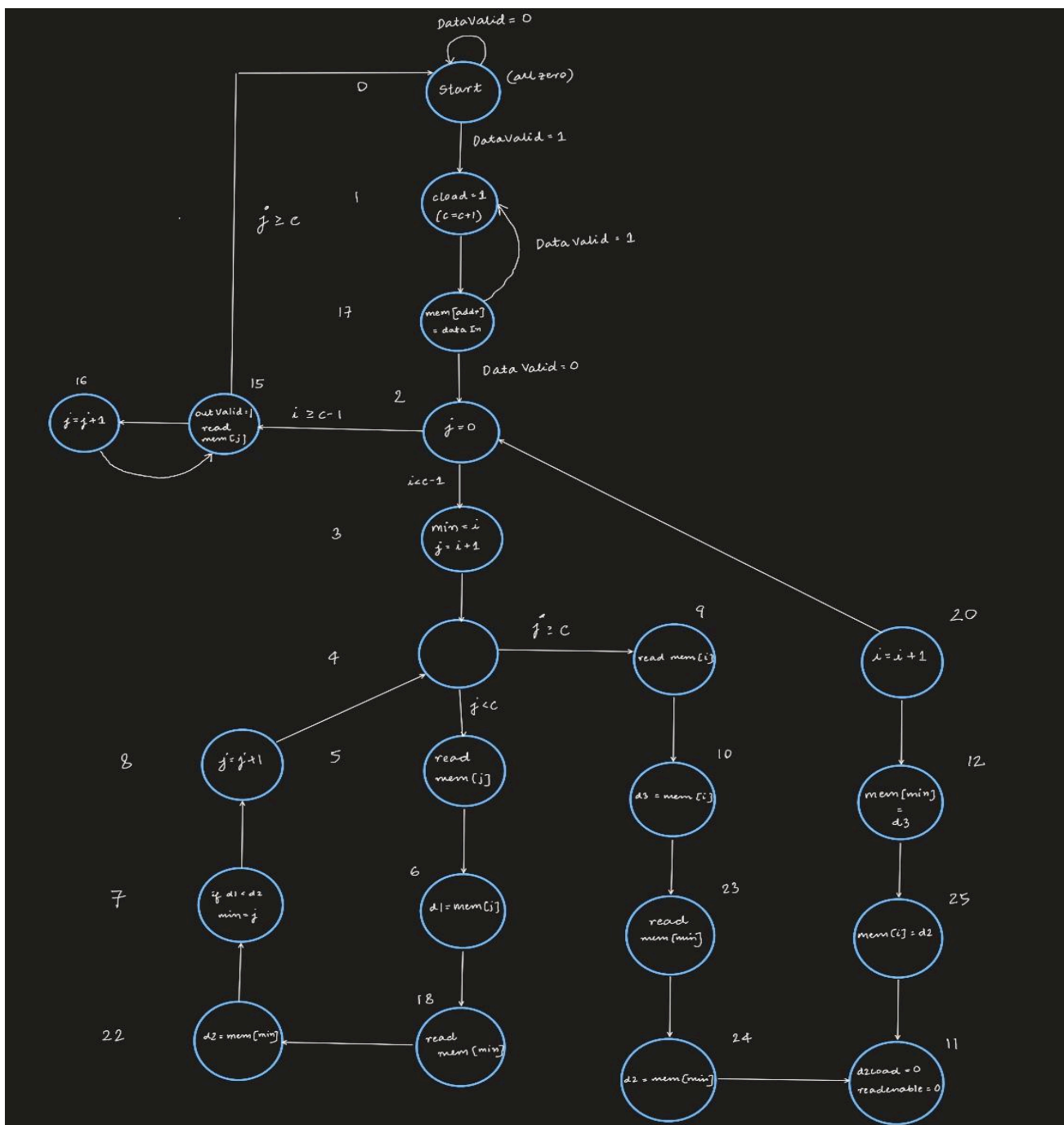
5. **Output Data Presentation:** Upon sorting completion, the FSM advances to a state for output presentation. Using a **datapath**, sorted values along with their memory addresses are sequentially outputted.

6. **Memory Implementation:** Leveraging a **single-port memory** module, the implementation ensures efficient data storage and retrieval. The entire system operates through a datapath, modifying **signals within the FSM** that interact with the datapath to effect changes.

By adopting a gate-level approach and utilizing a single-port memory, the design optimizes resource usage and maintains efficiency.

Through the careful manipulation of datapath signals within the FSM, the implementation achieves the desired functionality while adhering to the assignment constraints and objectives.

FSM



SIMULATION RESULTS

